Learning Efficient and Accurate Detectors With Dynamic Knowledge Distillation in Remote Sensing Imagery

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***Abstract*— Deep convolutional neural networks (CNNs) have brought a tremendous increase in detection accuracy, but too cumbersome model makes them hard to deploy on low computation edge devices, such as satellites and unmanned aerial vehicles. A promising method to tackle this problem is knowledge distillation (KD), which makes models lightweight with satisfactory accuracy. For remote sensing images, the objects are usually environment-related and located in a cluttered scene. The features that objects' semantic information relies on are tangled. However, existing distillation methods only imitate feature distribution derived from regions, including objects resulting in poor performance. Furthermore, masses of instances generated by teachers are blindly inherited, even if some of them are outliers. In this article, we propose a general and effective KD framework called dynamic knowledge distillation (DKD). First, our framework leverages the dynamic global distillation (GD) module to discover valuable regions from the foreground and background for multiscale features imitation, avoiding ignoring the potential geographical spatial relationship. Second, we propose a dynamic instance selection distillation (ISD) module to give students the ability of self-judgment through the magnitude of detection loss. Third, toward more accurate handling of hard samples in regression, a training-status-aware loss is tailored to guide students mine knowledge about objects with large aspect ratio or small size. Extensive experiments are conducted to show the effectiveness of DKD framework. The detection results on DOTA and NWPU VHR-10 dataset illustrate that our method is suitable for single-stage, two-stage and even anchor-free detectors. It shows the state-of-the-art performance. The code will be publicly available.**

***Index Terms*— Context information, instance selection, knowledge distillation (KD), object detection, training-status-aware weighted.**

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I. Introduction

RECENT years have seen great improvement in the accuracy of object detection, relying on deep convolutional neural networks (CNNs) [1]–[8]. Afterward, shortening the time from images acquisition to interpretation has gradually become the hotspots in the field of remote sensing. However, most state-of-the-art detectors are cumbersome to deploy on devices with low storage and low power consumption, such as satellite on-orbit processing, unmanned aerial vehicle (UAV) tracking, and automatic driving recognition. To address this problem, acquiring neural networks with light computation cost has attracted wide attention in most practical applications.

Hence, a series of technologies has been developed. Pruning [9], [10], quantization [11]–[13], and decomposition [14] directly compress the cumbersome network, effectively reducing computation and model size. The specific convolution structure [15]–[17] is also designed to obtain a lightweight detector. However, these compact networks may require customized hardware and software to realize practical acceleration. More importantly, there is always a certain gap of accuracy between compressed models and large models. To maintain the performance of compact detectors, knowledge distillation (KD) [18] is a promising high-level alternative. Rather than training from scratch, KD encourages lightweight student networks to achieves better performance by imitating the interclass structural knowledge [18]–[25] from complicated teacher networks.

KD has recently been put forward for detection tasks in natural scenes and has achieved impressive results. Wang *et al.* [26] proposed a fine-grained feature imitation that asks the student network to imitate the teacher network on the areas near the ground truth. Sun *et al.* [27] further used a 2-D Gaussian mask in the ground-truth regions for feature distillation. Li *et al.* [28] made a exploration by distilling the instances sampled by region proposal network (RPN). Nevertheless, the above distillation approaches are not suitable for remote sensing objects that are shown in Fig. 1. The contributing factor is that those KD methods follow a "static" mindset, which is embodied in the following.

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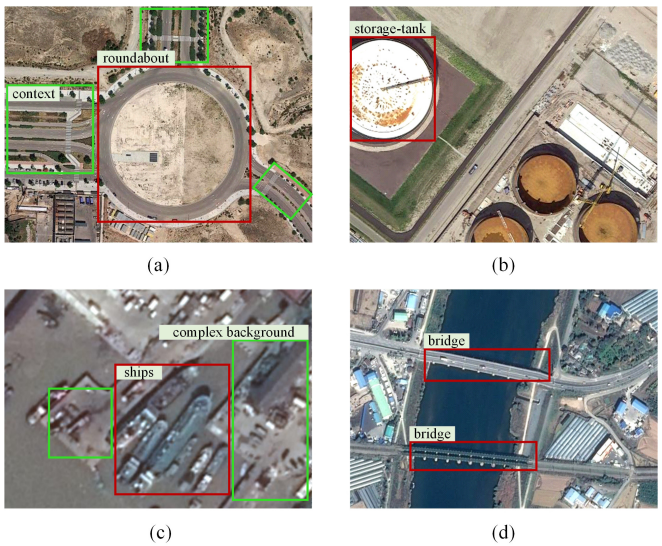


Fig. 1. Visualization of remote sensing images. (a) and (b) RA is generally surrounded by radial roads, while storage tank is not. Existing KD methods ignore the context information and easily cause confusion between them. (c) Ships appear in a cluttered complex scene, which can easily mislead the teacher network. (d) Bridge with extreme aspect ratios is difficult to regress in the student network.

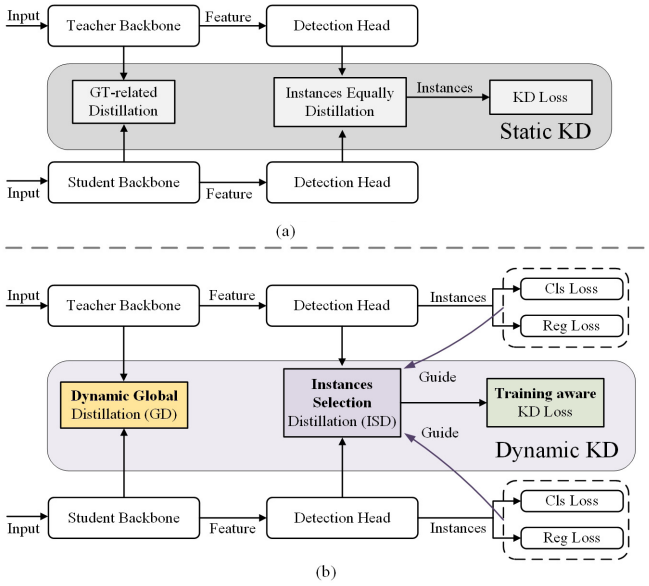


Fig. 2. Pipelines of KD in detection. (Top) Existing methods. (Bottom) Our proposed DKD. The improvement of DKD focuses on three points. (a) Valuable regions from the foreground and background are automatically aggregated and simulates in the backbone, avoiding the neglect of the context information (previous works). (b) DKD also dynamically selects the distilled instances in the detection head (ours). KD loss function is well designed, which will change dynamically with the difficulty of regression.

1) The distilled regions depend on fixed ground truths. As shown in Fig. 2(a), the existing methods only distill the GT-related regions while ignoring the background throughout the training process. In remote sensing scenes, objects are generally environment-related: the roundabout (RA) is generally surrounded by radial roads and the water is on both sides of the bridge. More importantly, recent research has shown that background information is also helpful for KD and visual recognition [29]–[32] in some cases. Consequently, it is necessary to automatically find valuable regions from the foreground and background, highlight distinguishing features, and avoid suppressing some contextual information.
2) The instances of distillation are indiscriminately and equally inherited. In the above approaches, the instances to be mimicked are fixed in the entire process of KD. However, objects for detection appear in complex and cluttered scenes, which can easily mislead the teacher network to make mistakes. Once the supervision signal provided by teacher is "wrong," students will directly update the gradient in the wrong direction. Therefore, students need to have the ability to self-judgment when and where to imitate. In addition, the general distillation paradigm treats all instances "equally," while many small and large aspect ratio objects specific to remote sensing are easily overlooked.

In response to those challenges, with few relevant studies in remote sensing field, we design a novel dynamic knowledge distillation (DKD) framework. DKD continuously extracts network status and attributes of remote sensing objects to guide distillation.

As shown in Fig. 2(b), the DKD is divided into three parts, which cooperates with each other to realize dynamic distillation. Dynamic global distillation (GD) module is the first part located in the backbone. Given the input image, the GD module performs multiscale feature distillation through a set of GD weights. It enables the network to automatically adjust the degrees of the contributions from the foreground and background pixels, depending on the content of the current image. Another part of our DKD is the dynamic instance selection distillation (ISD) module, which aims to deal with teacher' mistakes in teaching. The ISD allows the student detector to selectively imitate teacher prediction instances at the detection head level. On the basis of hard ground-truth supervision, this module additionally provides filtered structural information among different categories and regression. With the guidance of ISD module, the outputs of the student network containing objects are dynamically selected based on the magnitude of detection loss. To further improve accuracy, our framework introduces a training-status-aware loss based on the shortest edge. The commonality between small objects and objects with large aspect ratio is that the shortest edges of the prediction boxes are small. Hence, every time the gradient is updated, the loss function measures the difficulty of regression of the shortest edge and enables the student network to further dig and learn the hard instances of regression in the distillation process.

Specifically, the DKD framework is only introduced in the training phase and is cost-free during inference. By replacing the backbone from ResNet-152 [33] to ResNet-18, DKD can reduce the inference time and parameters of typical detectors by nearly three times and four times while keeping the same accuracy.

The contributions can be summarized as follows.

1) We present a general and effective DKD framework, which dynamically distills different positions of the detector according to the task priors. Furthermore, DKD has robust generalization for typical detectors.

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2) The dynamic GD module is proposed to comprehensively capture spatial importance for multiscale features distillation. A changing global mask is generated to avoid the confusion of objects caused by neglecting the surrounding information.
3) The dynamic ISD module is adopted to enable the student network to distinguish the teacher's wrong guidance and have the self-judgment ability. The distillation instances are dynamically selected based on the magnitude of detection loss.
4) Toward more accurate handling of hard samples in regression, a training-status-aware loss based on the shortest edge is tailored. It guides student to mine vital knowledge about objects with large aspect ratio or small size.

The remainder of this article is arranged as follows. Section II briefly introduces some related works. Section III explains the details of our method, and extensive experiments are shown in Section IV. Finally, the conclusion is outlined in Section V.

II. Related Works

In this section, we will briefly review related works from three aspects: object detection, deep network compression and acceleration, and KD in object detection.

A. Object Detection

At present, popular object detection models [1]–[6] are basically based on deep neural networks. According to the change process of proposals, it can be roughly divided into single-stage methods and two-stage methods, each of which has its own advantages in speed or accuracy. The two-stage algorithm follows "coarse-to-fine" [3]. It first focuses on finding out where objects appear and get proposals. Then, it classifies proposals along with returning to more accurate positions. In the two-stage algorithm, the RPN network uses anchor mechanism to link region generation with convolution networks. It achieves relatively high accuracy while consuming resources and time at the same time. For practical applications, smaller models and lower computational complexity are the key points. As pioneers of the single-stage detectors, SSD [1] and YOLO [2] achieved faster speed. They found the location of objects and predicted the category simultaneously. Single-level structure abandons the process of proposal, which is more suitable for practical application, but the introduction of category imbalance makes the recognition accuracy decline. Faced with this dilemma, RetinaNet [4] reduced the weight of massive amounts of simple negative samples in training by multiplying the common cross-entropy loss with a focal term. The above works are based on anchor mechanisms, while FCOS [5] was proposed as an anchor-free detection network. It avoided adjusting hyperparameters and calculations related to anchor boxes. By exploring the differences between RetinaNet and FCOS, ATSS [6] designed a method of selecting positive samples adaptively and realized the state-of-the-art performance in natural scenes without introducing extra overhead.

With the emergence of a series of excellent networks and high-resolution satellite (HRS) data, the object detection of remote sensing field has been further developed [7]–[9], [11], [12], [34]–[43]. However, unlike natural scenes, the unique complexity of remote sensing scenes leads to greater challenges. In order to effectively detect multiscale objects in high-resolution satellite images, Guo *et al.* [38] proposed an object proposal network and a detection network, sharing a multiscale basic network. SRAF-Net [7] was proposed to deal with irregular objects through a context-based deformable module and a multitask detection head. SCRDet++ [39] discussed the problem of noise and pointed out a novel InLD component. It approximately decoupled the features of different classes into their respective channels, thus preventing the interference of noise. In addition, remote sensing objects have directionality. R3Det [40] refined the single-stage detector to detecting rotating objects with a large aspect ratio and dense arrangement. FMSSD [8] was dedicated to solving the overwhelming small object problem. It designed an atrous spatial feature pyramid module, which improved the detection accuracy of small targets through a relatively fast single-stage algorithm.

B. Deep Network Compression and Acceleration

Although the speed of single-stage detection network is superior, these methods based on deep network are still equipped with large models and high computational complexity. They cannot meet the low delay requirements of terminal devices. In order to further accelerate the model, abundant techniques have been proposed. The first way to compact the model is to directly compress the cumbersome network. Quantization [11]–[13] mainly used low-bit representation to achieve effective compression, and network pruning [9], [10] focused on reducing redundant connections in large models. Low-rank factorization [34], [35] decomposed the large matrix to estimate the informative parameters. The second way is to design a special convolution structure, such as ShuffleNetV2 [17], MobileNetV3 [15], and SqueezeNet [44]. They effectively reduced the memory footprint during inference, which is specifically tailored for mobile devices. However, these methods may require customized hardware and software to achieve actual acceleration. In addition, there is a certain accuracy margin between the compact models and the cumbersome state-of-the-art models. To improve the compact detector's performance, KD is a promising alternative.

KD, proposed by Hinton [18], finds a sweet spot between speed and precision. Its key idea is to transfer knowledge from a cumbersome teacher model to a shallow student model, through learning interclass information [18]–[20] located in softmax output. Following the teacher–student algorithm, KD has been widely used in lightweight classification models. FitNet [20] explored the teacher's middle-level representation as a soft label, allowing student to imitate the teacher's full feature map. Sergey *et al.* [19] improved the performance of small models by mimicking the teacher's attention map at different stages after feature fusion. Yim *et al.* [45] chose to care more about the relationship between layers of the large

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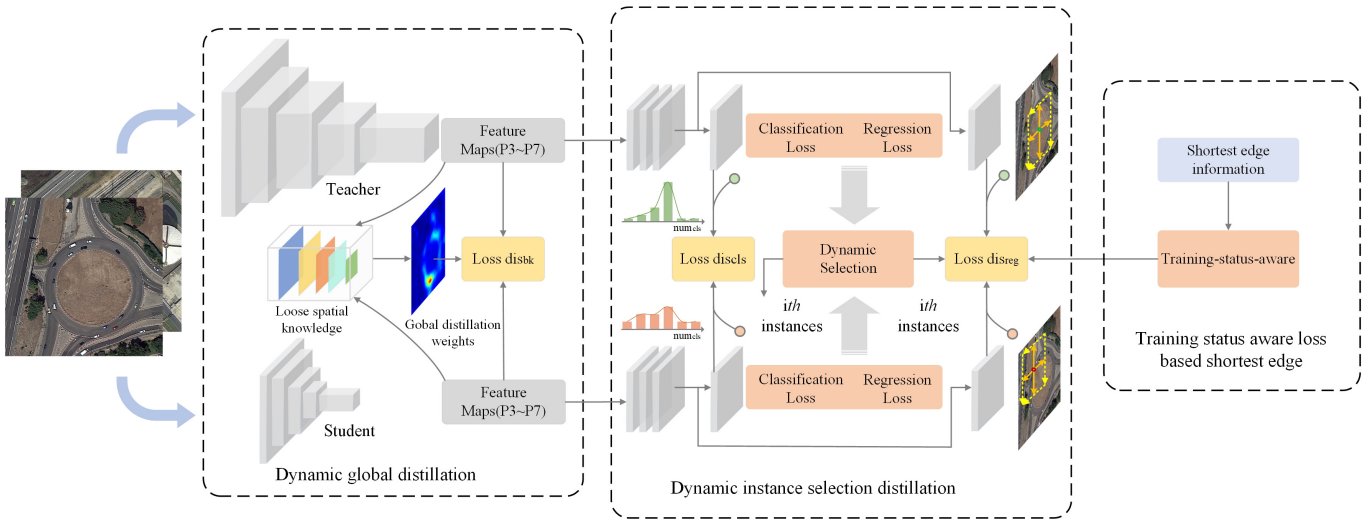


Fig. 3. Overall architecture of DKD framework. It contains the dynamic GD module and dynamic ISD module, as well as training-status-aware loss based on shortest edge. Among the framework, gray parts represent the original structure of the detection model, while others are the added distillation modules in the training phase. Yellow means the distillation loss to be optimized.

model, and Park *et al.* [46] proposed relational distillation to find the potential relationship between samples. A similar work has sprung up, but most of them focus on classification tasks.

C. KD in Object Detection

Recently, KD has been gradually applied to detection tasks to compress object detectors in natural scenes. Chen *et al.* [47] distilled the various positions of the detector, but the imitation of the entire feature maps ignored the imbalance between foreground pixels and background pixels, which may lead to suboptimal results. To get around this conundrum, Wang *et al.* [26] proposed a fine-grained feature imitation method, which made the detector care more about local regions near objects. According to this line of thinking, Sun *et al.* [27] added a 2-D Gaussian mask in the ground-truth regions for feature distillation, highlighting the centric foreground pixels. The above methods reduced the imbalance caused by negative samples at the cost of discarding the background region completely. However, recent studies have proved that the background regions also contains vital knowledge [31], [32]. More importantly, remote sensing objects are generally environment-related. The regions around the targets and the background area should be paid attention to during the distillation process.

Unlike the simple classification task, the detection task is more complex. It is necessary not only to distinguish the object category but also to perform reliable regression. Chen *et al.* [47] attached the teacher boundary regression loss to the regression component and completed the transmission of nondiscrete and borderless regression information, whereas there is no distinction between the objects with different degrees of difficulty in regression. Li *et al.* [28] used the $L2$ loss to punish all the output differences between student and teacher in the detection head. Sun *et al.* [27] used the binary cross entropy and $L1$ loss to simulate the positive samples of classification and regression heads. Nevertheless, remote sensing objects appear in complex and cluttered scenes, which can easily mislead the teacher network to make mistake. The above methods blindly imitated masses of instances generated by the teacher network, even if some of them are outliers.

In summary, the existing distillation methods ignore the important role of background, and the contextual information around the objects may be lost. In addition, the student network should not blindly follow the teacher network without self-awareness.

III. Methodology

The total structure of our proposed DKD framework is shown in Fig. 3. In this section, the overall architecture of DKD is illustrated in Section III-A. Sections III-B–III-D introduce the details of three dynamic distillation modules in DKD.

A. Problem Setup

KD transmits the information from a complicated teacher network to a compact student network while maintaining the strong performance. Define $Q = \{q_1, q_2, \ldots, q_n\}$ as all sample points to be mimicked on the feature map. As for KD, we hope that the output of the teacher network could offer a reasonable mild target for the student network to imitate and relieve the abrupt imitation to outliers. Hence, Q can be divided into two sets

$$Q = A \cup B \tag{1}$$

where A is the set containing reasonable samples and B is a set of outliers that may provide wrong guidance for student.

Generally, each position L of the detector can be distilled, such as the backbone network and the detection head. For the qth sample in A, the evaluation criteria for distillation at position l can be defined as $\psi_l^q(\theta)$. By adjusting the parameters θ of student network, $\psi_l^q(\theta)$ makes student's s_l^q and teacher's t_l^q as close as possible. The closer s_l^q and t_l^q,

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the greater the value of $\psi_l^q(\theta)$. As a result, the overall goal of KD is illuminated as (2), and λ_l denotes the contribution of distillation at l

$$\psi(\theta) = \underset{\theta}{\operatorname{argmax}} \sum_{q\in A}\sum_{l\in L} \psi_l^q(\theta)\lambda_l. \tag{2}$$

Furthermore, for a certain position l, define $\varepsilon_{l_j}^q$ as the deviation between paired student–teacher responses at the jth layer. Supposing that the deviation obeys the Gaussian distribution as $\varepsilon_{l_j}^q \sim N(0, \sigma^2)$, it is obvious to know

$$t_{l_j}^q = s_{l_{j-1}}^q \theta_{l_j} + \varepsilon_{l_j}^q \tag{3}$$

$$p\left(t_l^q \mid s_{l-1}^q, \theta_l\right) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{\left(s_{l-1}^q\theta_l - t_l^q\right)^2}{2\sigma^2}\right). \tag{4}$$

In (3), $s_{l_j}^q = s_{l_{j-1}}^q \theta_l$ and $s_{l_0}^q$ represents the input image. Assuming that the deviation $\varepsilon_{l_j}^q$ is generally independent and identically distributed, the task of distillation can be transformed into a maximum likelihood estimation problem as

$$\begin{aligned}\log \psi_{l_j}(\theta) &= \underset{\theta}{\operatorname{argmax}} \log \prod_{q\in A} p\left(t_{l_j}^q \mid s_{l_{j-1}}^q, \theta_{l_j}\right) \\ &= m \log \frac{1}{\sqrt{2\pi}\sigma^2} + \underset{\theta}{\operatorname{argmax}} \sum_{q\in A} -\frac{\left(s_{l_j-1}^q\theta_{l_j} - t_{l_j}^q\right)^2}{2\sigma^2}.\end{aligned} \tag{5}$$

Based on the above derivation, we can formulate the overall goal of KD based on the least-squares method to help student network perceive reasonable target as

$$\begin{aligned}&\underset{\theta}{\operatorname{argmin}} \sum_{q\in A}\sum_{l\in L}\sum_{l_j\in l} w_q\left(s_{l_j}^q - t_{l_j}^q\right)^2\lambda_l \\ &\qquad = \underset{\theta}{\operatorname{argmin}} \sum_{q\in A}\sum_{l\in L} w_q\left(s_l^q - t_l^q\right)^2\lambda_l\end{aligned} \tag{6}$$

where w_q is added so that students can further dynamically adjust the degree of attention to each sample point.

Based on the above representation, we propose a unified and effective distillation framework called DKD. As shown in Fig. 3, DKD is composed of dynamic GD, dynamic ISD module module and a training-status-aware loss based on the shortest edge. After using GD module in the backbone (l = bk), the student network automatically aggregates and simulates the global spatial importance of the five-layer feature pyramid network (FPN). The ISD module is located at the detection head (l = cls&reg) to deal with the teacher's misjudgment. ISD will compare the performance of students and teachers in the classification and regression tasks and dynamically selected the instances to be distilled. Finally, training-status-aware loss further improves the degree of attention to small or large aspect ratio objects. Details will be described in Section III-A.

B. Dynamic GD Module

Existing distillation methods open a new door for lightweight detection models by mimicking features suggested in the same region proposal [28], the object region [26], or within the ground truth [27]. However, it is not applicable to remote sensing objects that usually have a certain geographical spatial relationship. If we simply imitate the object itself, it can be predicted that a considerable amount of knowledge will be discarded. RAs may be confused with oil storage tanks (STs), and football fields may be indistinguishable from track fields. Compared to distillation based on fixed ground truths, it is necessary to automatically find valuable regions from the foreground and background to imitation.

To achieve this goal, we proposed a dynamic GD module. In fact, GD defines a distillation loss constraint, which can extract vital knowledge on the basis of selecting an appropriate location l. It constructs an ingenious global and constantly changing mask to transform knowledge so that students can approach the worthy knowledge transmitted by teachers as much as possible.

We obtain the multiscale features of the backbone network as knowledge. For the CNN-based detector, research indicates that the performance improvement is mainly attributed to the better backbone features. In order to make student gain more, imitating on the backbone is an intuitive idea. The feature of multiple backbone layers can also significantly improve the robustness of detecting multiscale objects.

Take RetinaNet [4] as an example, and we extract the output of FPN from level p_3–p_7, which are spatially smaller than the input image by factors of {4, 8, 16, 32, 64}

$$\begin{aligned}S_{l=bk} &= \{p_3, p_4, p_5, p_6, p_7\} \\ T_{l=bk} &= \text{frozen}\{p_3, p_4, p_5, p_6, p_7\}\end{aligned} \tag{7}$$

where frozen denotes the operation of freezing teacher's parameters, which is a common practice in KD. l = bk indicates that the location of distillation is in the backbone (bk). After aligning features of different scales, GD distillation loss can be simply defined as

$$L_{\text{dis}_{\text{bk}}} = \sum_i \left\| I_{\text{bk},i} \times \left(\frac{\rho_{\text{adap}}(s_i)}{\left\|\rho_{\text{adap}}(s_i)\right\|_2} - \frac{t_i}{\|t_i\|_2}\right)\right\|_p \tag{8}$$

where i represents the ith scale, p refers to normlization type (use $p = 2$ in the experiments), and ρ_{adap} is the adaptation function to adapt s_i to the same dimension as t_i. In addition, even if S and T have compatible features, forcing student to approximate teacher through an adaptation layer will lead to better gains. Here, we use a 3×3 convolutional layer.

$I_{\text{bk},i}$ is a foreground–background collaborative mask, which is used to dynamically adjust the contribution of teacher–student gap of foreground and background pixels in the distillation loss. To avoid the reduction of contextual information, the mask must be both global and loose.

Pooling operation, playing a significant role in computer vision, can obviously be qualified for this task. In the past, pooling was generally used to obtain the invariance of images, while our work adopts it to model the invariance of the knowledge between teachers and students. The $I_{\text{bk},i}$ form based on pooling has two advantages. First, it will highlight the features of strong discrimination, which is convenient for subsequent tasks of detection. Second, it not only imitates

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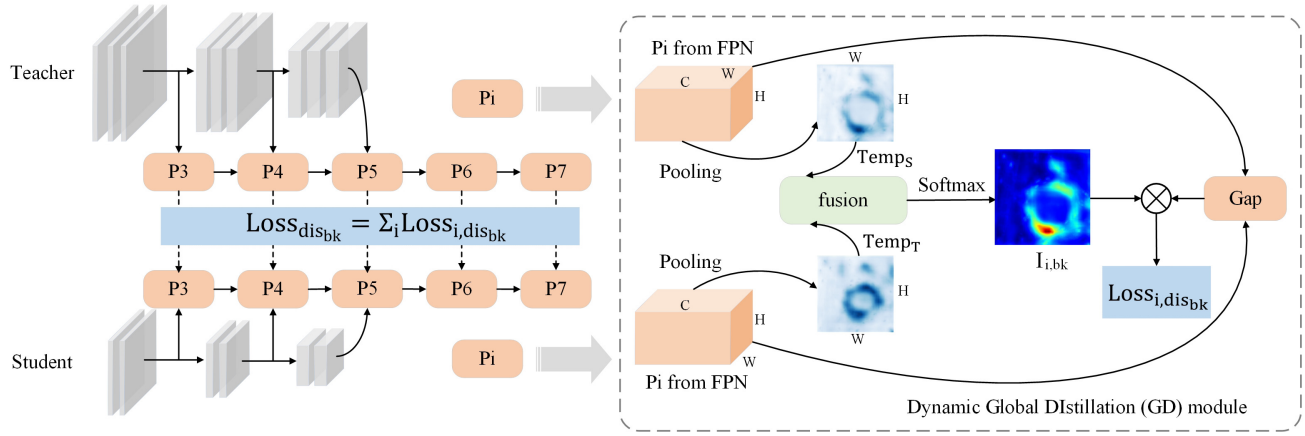


Fig. 4. Detail of the dynamic GD module. GD is able to capture global unique importance for multiscale features distillation and avoids the confusion of objects caused by forgetting the surrounding and even background information.

the object area but also emphasizes the knowledge of surrounding scenes that are geographically related to the object and conducive to identification. Specifically, it can be defined as

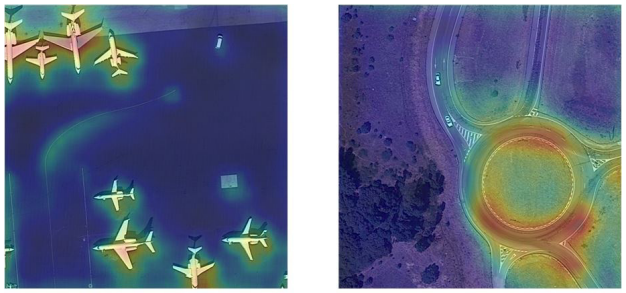
$$k_i = \frac{\text{pool}(s_i)}{\text{Temp}_S} + \frac{\text{pool}(t_i)}{\text{Temp}_T} \tag{9}$$

$$I_{\text{bk},i,\upsilon} = E \times \sum_g \frac{e^{k_i^\upsilon}}{e^{k_i^g}}. \tag{10}$$

In the field of KD, Hinton [18] believes that besides positive label, negative labels also carry key generalization ability. The above ideas coincide with the objectives of ours. For the detection task, in (9), we introduced Temp_S and Temp_T and used it to change the distribution of distillation weights $I_{\text{bk},i}$. The larger the Temp, the smoother the distribution of $I_{\text{bk},i}$, and the more attention will be paid to the beneficial background.

Unlike in the past, when $I_{\text{bk},i}$ is generated, we bring in information from both teacher $\text{pool}(t_i)$ and student $\text{pool}(s_i)$ to better integrate the status of the student itself. As s_i is constantly changing as the training progress, $I_{\text{bk},i}$ is also constantly changing mask. It enables the network to automatically adjust the degrees of the contributions from the foreground and background pixels, depending on the content of the current image. $I_{\text{bk},i}$ is calculated by softmax in the i dimension and then multiplied by appropriate normalization factor E. Specifically, (10) shows the calculation approach of the υth element on the mask $I_{\text{bk},i}$. The overall process of the GD module is shown in Fig. 4.

As shown in Fig. 5, we realized the dynamic GD through pool-based operations. However, in terms of specific implementation, different possible poolings, such as spatial, channel, horizontal, and vertical, determine different types of knowledge to be relaxed because the knowledge of the relationship between objects exists in the dimension of height and wide interaction. If it collapses, the mapping relationship will be lost. Information from other dimensions is not conducive to geospatial relationship modeling, so we only choose operation p_{spatial} and normalization factor $E = W \times H$. For more discussion, please refer to Section IV



(a) (b)

Fig. 5. Global mask generated by the GD module during feature distillation. (a) In addition to the plane itself, the feature responses of the runway line in the background area are also distilled. (b) Entrances, exits, and radial routes around the RA are also imitated.

$$\text{pool} = p_{\text{spatial}} \tag{11}$$

$$\text{pool}(s_i) = \sum_{c=1}^{C} s_l^{c,w,h}, \quad \text{pool}(t_i) = \sum_{c=1}^{C} t_l^{c,w,h}. \tag{12}$$

C. Dynamic ISD Module

The GD module has mined the appropriate global knowledge. However, it is far from enough because the object detection is a task of classifying and regressing objects simultaneously. One promising way is to add proprietary restrictions to those both tasks while combining the remote sensing properties.

Different from the general scenario, objects of remote sensing images appear in cluttered and complex scenes. Coupled with the influence of resolution and imaging factors, noisy information can easily mislead the network. Even teacher networks with more complex structures may suffer from poor regression accuracy or category misjudgment. Just like in real life, "no one is perfect." Although the teacher has seen a great deal of life, his comment cannot be right all along. In addition, the student will grow. With the training, the

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student's abilities will enhance. As the student progresses from the "kindergarten" level to the "graduate" level, he will be able to identify and question the teacher' opinions and study selectively.

Flow this intention, we design a mechanism called "dynamic ISD," which considers student's own growth and dynamically selects instances to be imitated. It acts on the end of the classification head and regression head (l = detection head). a_1–a_5 represent the output of the detection head in different scales

$$\begin{aligned} S_{l=\text{head}} &= \text{frozen}\{a_1, a_2, a_3, a_4, a_5\} \\ T_{l=\text{head}} &= \{a_1, a_2, a_3, a_4, a_5\} \\ \text{head} &\in \{\text{cls}, \text{reg}\}. \end{aligned} \tag{13}$$

For the convenience of description, we introduce the imitation details of the classification head and the regression head respectively.

1) Dynamic Instance Selection in Classification: The overall distillation process of classification head is as follows: first, the instances containing the objects are selected preliminarily and then refined dynamically according to the performance of teachers and students in the classification task.

Assuming that the detection network predicts N instance boxes, then the output values of the student and teacher classification headers, $s_i \in S_{l=\text{cls head}}$ and $t_i \in T_{l=\text{cls head}}$, represent the confidence of N prediction boxes in num_{cls} categories, with the size of $N \times \text{num}_{\text{cls}}$. We minimize the following forms of distillation loss for student-to-teacher imitation in classification head:

$$L_{\text{dis}_{\text{cls}}} = \frac{1}{N_{\text{cls}}} \sum_{i=1}^{N} \left[M_{\text{cls}} \times \left(\rho_{\text{adap}}(s_i) - t_i \right) \right]^2. \tag{14}$$

In $L_{\text{dis}_{\text{cls}}}$, ρ_{adap} means 1×1 convolutional layers, N_{cls} is a normalization coefficient, and the calculation method will be given later. M_{cls} is the key to promote effective distillation and student's independent judgment ability. It is a coefficient matrix consisting of 0 and 1 with the length of N. 0 means neglecting the current prediction box and 1 indicates learning. As shown in Fig. 6, the value of M_{cls} is determined by two factors: 1) intersection of union (IoU) of prediction box to ground-truth box and 2) the current student and teacher network classification loss values.

For generating appropriate M_{cls}, we first screen out the instances containing objects in the classification head. The factors can be boiled down to two points. First, the beneficial background features have been sufficiently imitated in the GD module. Second, the output of detection head generally has masses of negative samples [4]–[6], which may lead to suboptimal results. When the IoU between predicted boundary boxes and ground-truth boxes is high, we believe that those boxes are more reliable, which are worth learning. On this basis, we compare the classification loss $L_{\text{gt}_{\text{cls}}}$ between students and teachers on predicted boundary boxes and select the distillation instances dynamically. Suppose that Box_i has passed the preliminary screening, we obtain the loss function L_i, which represents the classification loss of the Box_i on num_{cls} categories

$$\begin{aligned} L_{\text{gt}_{\text{cls}}} &= \{L_1, L_2, \ldots, L_N\} \\ L_i &= \left\{ l_1, l_2, \ldots, l_{\text{num}_{\text{cls}}} \right\}, \quad i \in \{1, 2, \ldots, N\}. \end{aligned} \tag{15}$$

In the process of training, the changing loss value actually indicates the present ability of the network to distinguish the object, and we can use it as the basis to dynamically judge the level of students and teachers. In fact, rather than considering the loss of a bounding box in each category as num_{cls} independent criteria, our intuition is to treat it as a minimum unit. For a certain Box_i, its losses in all categories are summed up to obtain the discriminant factor J_i

$$\begin{aligned} J_{S,i} &= \sum_{k=1}^{\text{num}_{\text{cls}}} l_{S,i} \\ J_{T,i} &= \sum_{k=1}^{\text{num}_{\text{cls}}} l_{T,i}. \end{aligned} \tag{16}$$

Algorithm 1 Generator Matrix M_{cls}

Input: The classification result of student network in multiple scale $\text{SP}_{cls} = \{\text{SP}_{cls1}, \cdots, SP_{clsN}\}$, the student's prediction box in multiple scale before nms $Box = \{Box_1, \cdots, Box_N\}$, the ground truth of current image $G = \{g_1, \cdots, g_{num_{gt}}\}$, threshold ϕ, the loss of student and teacher on classification tasks $L_{stu,gt_{cls}}$ and $L_{tea,gt_{cls}}$.

Output: The coefficient matrix M_{cls}.

```
Initializes M_cls to a full zero matrix with the size of N × num_cls
Step1: Screen out the instances containing enough foreground
for g_i ∈ {g_1, ··· , g_num_gt} do
    for Box_j ∈ {Box_1, ··· , Box_N} do
        Q_i ← record the maximum IoU(Box, g_i)
    end for
end for
for g_i ∈ {g_1, ··· , g_num_gt} do
    for Box_j ∈ {Box_1, ··· , Box_N} do
        if IoU(Box, g_i) > 0.5 × Q_i then
            Set the corresponding position i in M_cls to 1
        end if
    end for
end for
Step2: Endow students with independent judgment
for SP_cls ∈ {SP_cls1, ··· , SP_clsN} do
    Calculate the discriminant factor J_{S,i} and J_{T,i} respectively based on L_{stu,gt_cls} and L_{tea,gt_cls}
    if J_{S,i} − J_{T,i} < φ then
    end if
end for
```

The smaller J_i is, the stronger the classification ability of the network has. When the discriminant factor $J_{S,i}$ of students is small enough compared with $J_{T,i}$, it indicates that the student network can no longer emulate teachers by distillation

$$J_{S,i} - J_{T,i} < \varphi \tag{17}$$

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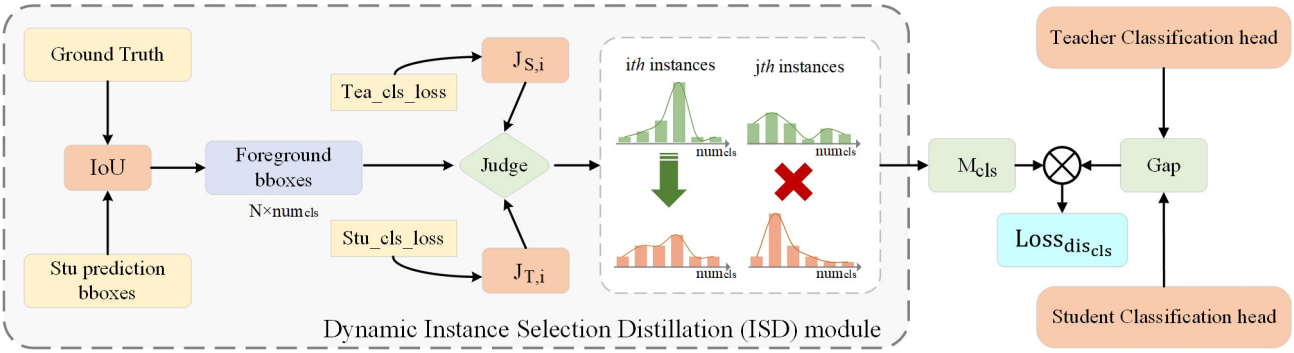


Fig. 6. Illustration of the dynamic ISD module in classification head. When the discriminant factor ($J_{S,i}$ and $J_{T,i}$) indicates that the teacher is relatively correct in the category probability distribution of the ith instance, the instance at the corresponding position of the student will imitate the predicted value from the teacher. On the contrary, the student will stick to its own judgment and not imitate.

where φ is a threshold factor. When $\varphi = 0$, $J_{S,i} < J_{T,i}$ means that teacher makes mistakes. However, the condition of 0 is too harsh. Let us take a smaller negative number as a relatively loose threshold. In the experiment, $\varphi = -1$, a better result will be obtained.

In summary, as shown in Algorithm 1, if the Box_i instance contains more than half of the objects area and has enough learning ability, the corresponding position of M_{cls} is set to 1; otherwise, it is set to 0. Correspondingly, the normalized factor in (18) is the total number of "1" in M_{cls}

$$N_{\text{cls}} = \sum_{i=1}^{N} \text{M}_{\text{cls}}(i). \tag{18}$$

2) Dynamic Instance Selection in Regression: For the distillation of regression head, the overall process is similar to that of classification. First, the positive samples output by the regression head is screened out, and then, the final instances are selected according to the teacher–student regression loss. The distilled regression head loss can be summarized as

$$L_{\text{dis}_{\text{reg}}} = \frac{1}{N_{\text{reg}}} \sum_{i=1}^{N} \text{M}_{\text{reg}} \times \text{dis}\left(\text{box}_{S_i}, \text{box}_{T_i}\right) \tag{19}$$

$$\text{dis}\left(\text{box}_{S_i}, \text{box}_{T_i}\right) = \sum_{j=1}^{4} \text{abs}\left(x_{S,j} - x_{T,j}\right) \tag{20}$$

$$N_{\text{reg}} = \sum_{i=1}^{N} M_{\text{reg}}(i) \tag{21}$$

where $\text{dis}(\text{box}_{S_i}, \text{box}_{T_i})$ represents the difference between the coordinates of the top-left corner (x, y) and the bottom-right corner (x', y') of the predicted bounding box for teachers and students and N_{reg} is the normalization coefficient. Here, M_{reg} is also a coefficient matrix of 0 and 1, and the process to generate the inner value is the same as M_{cls}.

D. Training Status-Aware Loss Based on the Shortest Edge

The detection of remote sensing scene typically commences with a training set consisting of objects with small size and large aspect ratio. In the regression task, these objects with domain characteristics are usually regarded as "hard cases." The underlying reason is that their width and height generally differ greatly from the default anchor, in which case it is difficult for the network to accurately predict the offset of the inherent anchor to the target position. Compared with the objects in natural scenes, the vast majority of objects with small size and large aspect ratio are a Gordian knot. However, it is also a property that can be more utilized. In the previous distillation methods, distillation obviously did not pay attention to data itself but treated all objects equally. This undifferentiated distillation method is bound to be detrimental to remote sensing scenes with a large number of hard cases. To address the above problems, we introduce a training-status-aware distillation loss function based on the shortest edge. Under its constraints, the network will automatically obtain the regression state of the student network to the instances. It is worth noting that the loss function we proposed considers not only the regression difficulty of the current object but also the current training state of the student network, which is constantly changing. The training-status-aware distillation loss $L_{\text{training-aware}}$ can be described as

$$L_{\text{training-aware}}(\text{bbox}_S, \text{bbox}_T) = \sum_i w_i L_{\text{dis}_{\text{reg}}}\left(\text{bbox}_{S_i}, \text{bbox}_{T_i}\right) \tag{22}$$

where i represents the ith box of the current network prediction and w_i is the dynamic weight. The localization task can be considered as regressing a boundary box from an anchor box to a nearby ground-truth box. Thus, the model needs to predict the location of each object, namely, five quantities: category, central point coordinates x and y, width w, and height h. With the anchor prior, the network regression results can be described as four offsets t_x, t_y, t_w, and t_h

$$t_x = \frac{\text{pre}_x - w_{\text{anchor}}}{w_{\text{anchor}}}, \quad t_y = \frac{\text{pre}_y - h_{\text{anchor}}}{h_{\text{anchor}}} \tag{23}$$

$$t_w = \log\left(\frac{\text{pre}_w}{w_{\text{anchor}}}\right), \quad t_h = \log\left(\frac{\text{pre}_h}{h_{\text{anchor}}}\right). \tag{24}$$

It can be seen from (24) that the width-to-height offsets t_w and t_h are logarithmic, which further limits the range of offsets and facilitates prediction. The absolute values of t_w and t_h indicate the difficulty of regression from the fixed anchor to the prediction box.

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For small objects and objects with large aspect ratio that are difficult to regress, we extract their commonness: the shorter side of the object's width and height is usually small. If we combine this idea with the expressions t_w and t_h, we will obtain the larger values of abs(t_w) and abs(t_w) as a measure of regression difficulty flag, which represents the ease of regression of the student's current bounding box

$$\text{flag} = \max(\text{abs}(t_w), \text{abs}(t_h)). \tag{25}$$

Obviously, during the training process, the predicted values of pre_w and pre_h for the width and height of the detection box by the student network are constantly changing, and the measurement value of regression difficulty flag will also be in a constant state of change. We expect that flag can direct the variation of instance weight w_i in (22), so as to construct a loss function that dynamically focuses on difficult regression instances.

Therefore, we design a function that monotonically increases with the difficulty degree of localization, and its value is always greater than 1. We use the exponential function as the main function. If the regression difficulty of the student network to the current instance is large, we increase the learning weight of this bounding box to the teacher in the distillation process. w_i is defined as

$$w_i = 1 + e^{-\frac{1}{\text{flag}_i}}. \tag{26}$$

For the box with index i in the student network, flag_i represents the localization difficulty of this box. When w_i is added to the loss function of the distillation regression head, the hard instances with a small size or extreme aspect ratio in the remote sensing images will have a greater impact on the gradient updating process, and this influence will undulate with the training status. The overall process of generating w_i is summarized in Fig. 7.

If we put the implementation details of w_i into the total distillation loss of the regression head, the shortest side weighted loss function of training-status-aware $L_{\text{training-aware}}$ can be described as

$$L_{\text{training-aware}}(\text{bbox}_S, \text{bbox}_T) = \sum_i \left(1 + e^{-\frac{1}{\text{flag}_i}}\right) L_{\text{dis}_{\text{reg}}}(\text{bbox}_{S_i}, \text{bbox}_{T_i}). \tag{27}$$

In summary, the ultimate goal of the student model is to minimize both the object detection loss and the total distillation loss. Integrating the above distillation terms produces our overall training targets of the student network, which can be expressed as

$$\text{loss} = L_{\text{detection}} + \sum_l L_{\text{dis}_l} \tag{28}$$

$$\sum_l L_{\text{dis}_l} = \lambda_1 L_{\text{dis}_{\text{bk}}} + \lambda_2 L_{\text{dis}_{\text{cls}}} + \lambda_3 L_{\text{training-aware}} \tag{29}$$

where λ_1, λ_2, and λ_3 are the balanced parameters for the overall distillation loss.

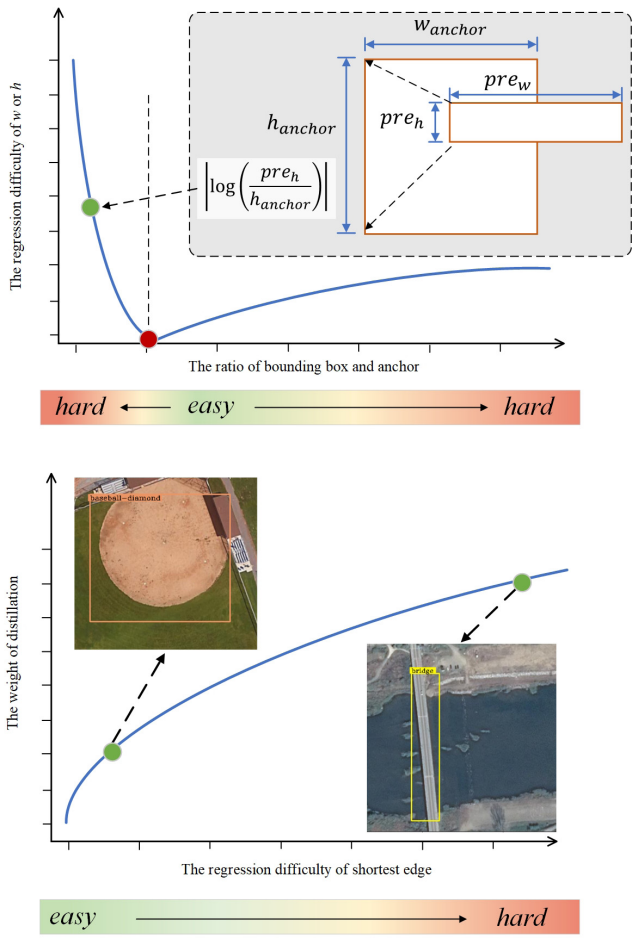


Fig. 7. Overall process of generating w_i with a value between 1 and 2, which assigns different importance to different scales and shapes during distillation.

IV. Experimental Results

This section evaluates our DKD framework and provides a concise description of the experimental details, such as datasets, evaluation metrics, parameter settings, and experimental results.

A. Datasets

1) DOTA Dataset: To demonstrate the effectiveness of our DKD framework, experiments are performed on the open-public DOTA dataset [48]. This article focuses on exploring the horizontal bounding box (HBB) detection task. DOTA provides a large-scale image dataset in the remote sensing field. It covers 188 282 instances over 15 categories, including plane, soccer ball field (SBF), bridge, baseball diamond (BD), small vehicle (SV), ground field track (GTF), large vehicle (LV), basketball court (BC), ST, RA, tennis court (TC), swimming pool (SP), harbor, and helicopter (HC). The pixel size of each image is between 800 × 800 and 4000 × 4000, including objects with different scales, directions, and shapes. Randomly select half of the original images as the training set, 1/6 as the verification set, and 1/3 as the testing set.

In this article, we crop the images to 1024 × 1024 pixels, overlapping 200 pixels, and simply expanded the categories with fewer objects. Finally, we acquire 15 749 patches in the

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training set, 5297 patches in the verification set, and 10 833 patches in the testing set.

2) NWPU VHR-10 Dataset: The NWPU dataset is a geographic remote sensing dataset for object detection. The images are cropped from the Google Earth and Vaihingen datasets and then manually annotated by the experts. It has 650 images of objects and 150 background images for a total of 800 in ten categories, including airplane, TC, ship, ST, BC, harbor, BD, ground track field, bridge, and vehicle. We randomly select 60% of the images as the training set, 20% as the verification set, and 20% as the testing set.

B. Evaluation Metrics

1) Measure the Accuracy of the Model:

a) Intersection of union: For the detection model with unstructured output, we can evaluate the results from the fitting degree between the detection boxes and the ground truths. The IoU is defined as the ratio of the intersection and union of two boxes, generally used to quantify the fitting degree. During evaluation, according to the IoU of predicted boxes and ground truths, each sample will be assigned attributes: true positive (TP) for correctly matching, false positive (FP) for wrongly predicting the background as an object, and false negative (FN) for undetected object.

b) Recall and precision: To evaluate the model, all the detection boxes are sorted in order of confidence score from high to low and then traversed. In the traversal process, the recall rate and accuracy rate can be calculated through the attributes of the current sample

$$\text{Precision} = \frac{\text{N}_{\text{TP}}}{\text{N}_{\text{TP}} + \text{N}_{\text{FP}}} \tag{30}$$

$$\text{Recall} = \frac{\text{N}_{\text{TP}}}{\text{N}_{\text{TP}} + \text{N}_{\text{FN}}}. \tag{31}$$

c) Mean average precision: By recording the precision and recall (P-R) of each step, P-R curves for each category can be plotted. In order to comprehensively consider the accuracy under different recall rates, we use the area of P-R curve to evaluate the model $\text{AP} = \int_0^1 \text{PdR}$. AP is the area between the curve and the coordinate axis. Mean average precision (mAP) is the average of APs for all classes.

2) Measure the Efficiency of the Model: The following conditions hold. *Frames Per Second (FPS):* It is the number of frames detected per second, which represents the running speed of the deep learning model. We input 1024×1024 pixel resolution slices of the entire test set and calculate the average number of images processed per second as FPS.

Parameters: It is the number of network parameters, representing the size of the model.

Floating Point Operation (FLOP): This measures the amount of computation and the complexity of the model.

C. Parameter Settings

In distillation, the DKD framework is applied to the object detection model. It is worth noting that the distillation frame only works during training and is cost-free during inference. Our code is based on MMDetection [49]. Since DKD is customized for remote sensing imagery, the overall parameter settings of the detector mostly follow AerialDetection [50]. In the process of training distilled student network, tricks, such as multiscale training and testing, are not used.

During the training process, the network uses stochastic gradient descent (SGD) to train the model with 24 epochs, each minibatch containing four images. The initial learning rate is 0.0025, which decreases by a factor of 10 in the 16th and 22th epochs. The weight decay is set as 0.0001 and the momentum is 0.9. The detector receives the input image with 1024×1024 pixels and is trained on the training set and validation set together. The backbone network is pretrained in ImageNet. All experiments are carried out on an RTX GPU, with CUDA 10.1.

For the DOTA dataset, the detector processes each patch in the inference. Then, the output of patches is merged to build the detection result of the whole image. In addition, NMS filters the overlapping boxes through the IoU threshold value of 0.3.

D. Evaluation of Our Model

1) Baseline Models: Our method is eager to solve the application problem of detection network in the terminal scene with low delay requirement, which needs smaller volume and faster speed on the premise of ensuring accuracy. We evaluate our DKD framework based on current typical detectors. For two-stage detectors, a widely used Faster R-CNN [3] is chosen, and for single-stage detectors, we mainly measure the performance using RetinaNet [4] and ATSS [6]. For anchor-free detectors, FCOS [5] is adopted. Following the general paradigm of teacher–student selection in the detection task, the differences between teacher and student are reflected in backbone networks of different sizes, such as ResNet [33] networks with depths of 101 and 18. In the experiment, we mainly use RetinaNet with the backbone of ResNet-101 as the teacher network and RetinaNet with the backbone of ResNet-18 as the student. The former is labeled as "Teacher-res101," and the latter is "Student-res18." In particular, we find that student's performance is positively correlated with teachers' scores in the detection task. Therefore, "Teacher-res101" is not raw, and it adds multiscale training to achieve a higher teacher performance.

2) Ablation Experiment: We first carry out experiments to explore the contribution of each distillation module to the final performance and the robustness of our method to different parameters. Finally, we combine these modules to make joint analysis.

The ablation experiments are based on RetinaNet and conducted on the DOTA dataset. The test result was obtained by submitting the prediction to the official DOTA website. The evaluation server only returned mAP. Table I shows the experimental results of each module. Teacher and student bring 72.42% and 63.69% accuracies, respectively, with a gap of 8.73%.

a) Effect of dynamic GD module: The GD module located in the backbone brings about the improvement of 6.55% mAP. Especially, the promotion is more obvious for

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TABLE I
COMPARISON OF THE PERFORMANCE OF EACH DETECTION METHOD ON THE DOTA DATASET

Method	Plane	BD	Bridge	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	Harbor	SP	HC	mAP
Teacher-res101	88.91	83.10	51.57	62.59	72.48	72.57	78.32	90.39	80.91	79.78	53.75	67.77	74.44	71.43	58.37	72.42
Student-res18	86.64	71.61	39.19	51.27	71.64	64.52	77.02	89.66	66.61	71.74	40.55	55.38	63.76	69.02	36.80	63.69
ISD_cls	87.97	76.87	47.01	60.22	74.19	65.00	78.29	90.76	76.37	74.61	48.14	66.49	70.08	74.37	45.27	69.04
ISD_reg	88.92	78.42	46.62	62.31	73.07	63.79	76.48	90.86	82.25	74.91	50.63	59.76	71.42	71.74	48.79	69.33
ISD	88.56	81.15	45.30	62.89	73.11	64.44	77.28	90.80	77.39	76.41	61.65	62.33	71.37	73.14	51.81	69.84
GD	88.67	79.46	47.53	58.91	73.64	65.30	78.51	90.78	77.19	76.61	53.48	65.01	72.51	73.39	52.69	70.24
GD + ISD	89.15	78.03	47.62	62.32	74.20	65.32	78.25	90.69	76.85	75.77	55.19	66.36	72.16	72.11	55.13	70.61
GD+ISD+Training-aware	89.48	81.48	46.38	60.52	76.25	64.18	78.36	90.79	78.60	78.31	53.12	65.06	73.05	74.11	59.98	71.31

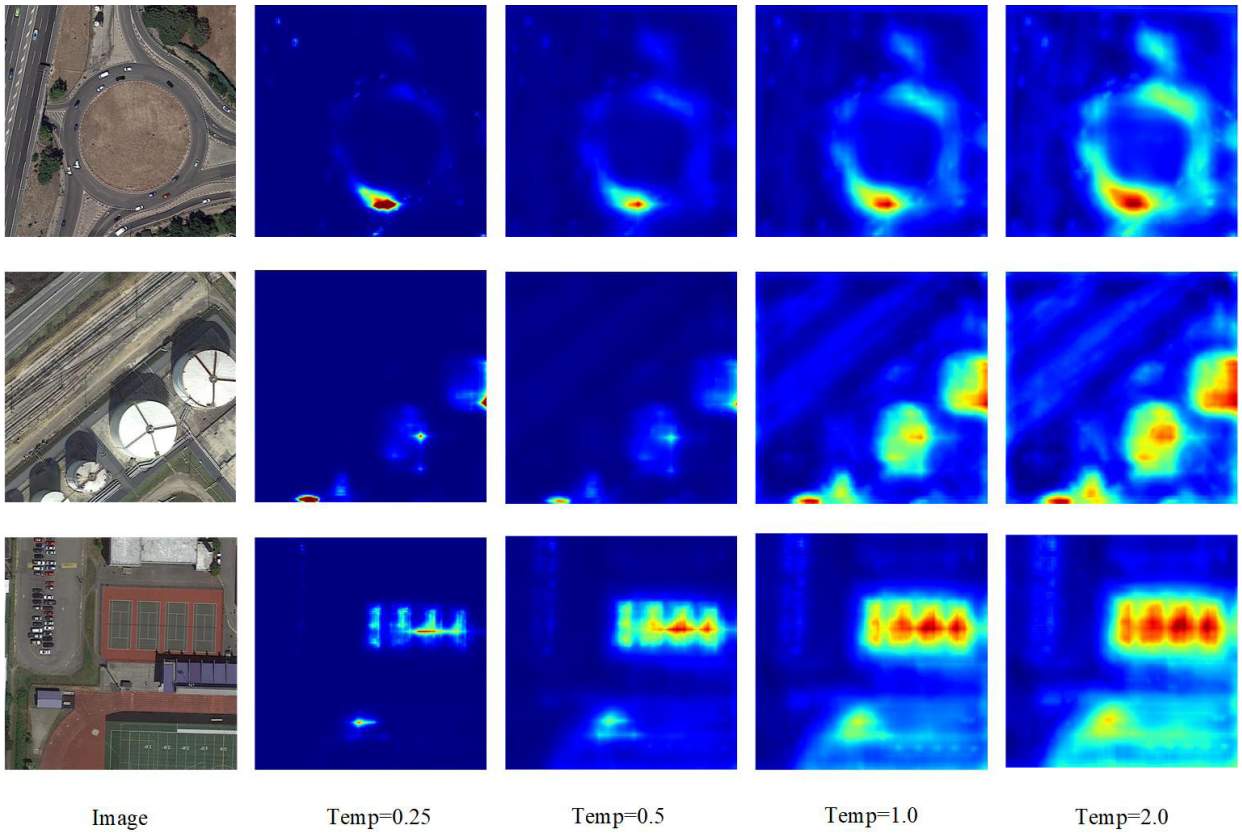


Fig. 8. Visualization of the GD weight mask I_{bk} at different temperature coefficients on the DOTA dataset.

the objects whose geographical spatial relationship can help distinguish. The confusing RA and ST are promoted by 9.63% and 4.87% mAP, respectively; ground-track-field, football field, BC, and other venues have all increased by more than 8% mAP. The bridge and HC are also upgraded by 9.34% and 15.89%, respectively.

Fig. 8 shows the influence of different temperature coefficients on the GD mask I_{bk}. For the detection task, we introduce the temperature coefficient T to change the distribution of 2-D mask I_{bk}. As shown in Figs. 8 and 9, the smaller t is, the more attention is paid to the pixels with higher attention in knowledge extraction, and the sharper the data distribution is. The larger the temperature coefficient, the smoother the attention distribution in I_{bk}. In addition to the object itself, the information surrounding its spatial position is also concerned to some extent. For example, the radial roads around the RA are also learned during distillation.

Table II shows the influence of different temperature coefficients on the results. It serves to show that the worst hyperparameter will only cause the final performance of the model to decrease by 0.26% mAP compared with the highest mAP, but it is still 6.29% mAP higher than that of the baseline. This proves that our method is robust and not sensitive to hyperparameter.

The GD module needs a relatively loose GD mode, and pooling can assist its implementation. Table III shows the influence of different pooling methods to generate I_{bk}. We have pooled four dimensions: space, channel, horizontal, and vertical. Although they are all "loose" knowledge, different relaxed dimensions actually determine different kinds of knowledge.

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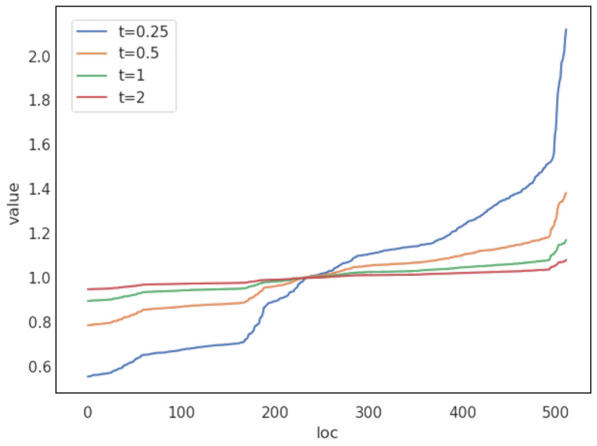


Fig. 9. GD weight I_{bk} distribution under different temperature coefficients on the DOTA dataset. It show that the smaller T, the flatter the distribution, and the larger T, the steeper the distribution.

TABLE II

INFLUENCE OF DIFFERENT TEMPERATURE COEFFICIENTS T ON THE DOTA DATASET

$Temp_s = Temp_t$	0.25	0.5	1	2
mAP	70.00	70.01	70.24	69.98

TABLE III

INFLUENCE OF DIFFERENT POOLING DIMENSIONS ON EVALUATION RESULTS ON THE DOTA DATASET

	$P_{spatial}$	$P_{channel}$	$P_{horizontal}$	$P_{vertical}$	mAP	ΔmAP50
Teacher-res101					72.82	
Student-res18					62.69	
Dis-res18	✓				70.24	+7.55
		✓			69.89	+7.20
			✓		68.02	+5.33
				✓	69.09	+6.40
	✓	✓			69.04	+6.35
	✓		✓	✓	57.12	-5.57

The best effect is achieved in space alone, which is consistent with the problems we need to solve. In order to preserve the knowledge of geospatial relations between objects, the information of the dimensions, where W and H interact on the feature map, cannot be lost. For our task, the collapsed spatial dimension will lead to bad results, while the supervision of other dimensions will introduce extra noise and lead to poor results. From an effect point of view, both 1×1 convolution and P_{spatial} pooling can generate knowledge of spatial dimension for aggregation. However, the frozen teacher network prevents the convolution of the teacher part from updating the parameters, so it cannot be used for comparison.

b) Effect of dynamic ISD module: The ISD module located in the detection head brings 6.15% mAP to the student model by dynamically selecting distillation instances. When acting separately on the classification header and regression header, the 5.35% and 5.64% mAP of the student model are promoted, respectively. Actually, the generation of M_{cls} and M_{reg} includes two steps. The first step is to collect prediction boxes containing enough positive instances on multiple scales, and the second step is to decide whether to learn the selected instances through the student's self-judgment. In order to prove that the above two steps are correct and effective, we explored as shown in Table IV, and the suffix "dy" indicates whether to introduce dynamic selection. The results show that they are helpful to students and give them the motivation to make independent judgments so that students can go up a flight of stairs.

TABLE IV

VALIDITY OF EACH PART OF THE ISD MODULE ON THE DOTA DATASET

	cls	cls_dy	reg	reg_dy	mAP	ΔmAP
Teacher-res101					72.42	
Student-res18					63.69	
Dis-res18	✓				68.29	+4.60
	✓	✓			69.04	+5.35
			✓		69.16	+5.47
			✓	✓	69.33	+5.64
	✓	✓	✓	✓	69.84	+6.15

TABLE V

HYPERPARAMETER ANALYSIS OF DISCRIMINANT THRESHOLD ϕ ON THE DOTA DATASET

Dynamic threshold	-0.5	-1	-2	-4
mAP	68.73	69.04	68.53	68.29

TABLE VI

ARGUMENT OF THE WAY OF WEIGHT ON THE DOTA DATASET

Method	mAP	ΔmAP
$2\text{-}flag_i$	70.8	+0.2
$e^{-flag_i} - 1$	71.3	+0.7

TABLE VII

EFFICIENCY OF RETINANET SERIES NETWORKS BEFORE AND AFTER DKD ON THE DOTA DATASET

Method	mAP50	Flops(G)	Params(M)	FPS(img/s)
Retina152	69.19	371.34	71.03	11
Retina101	68.72	293.4	55.39	15
Retina18	62.69	163.7	19.9	28
DKD_Retina18	71.3	163.7	19.9	28

Fig. 10 proves the effectiveness of the ISD module from a qualitative point of view. The top of the prediction box is labeled with the prediction category and confidence score. Only confidence values greater than 0.3 will be displayed. Take the first line as an example. Compared with Fig. 10(a) and (b), the common distillation method gives an improvement in the confidence of the SP and inhibited the misjudgment of the central road. However, distilled [Fig. 10(b)] without dynamic selection did not recognize the cars scattered in the image. After exploring the reasons, we find that for this image, the teacher has lower confidence on the car instance compared to student [Fig. 10(a)]. This shows that for some complex scenes under remote sensing, when the teacher is unable to make a wrong judgment, its guidance will bring negative

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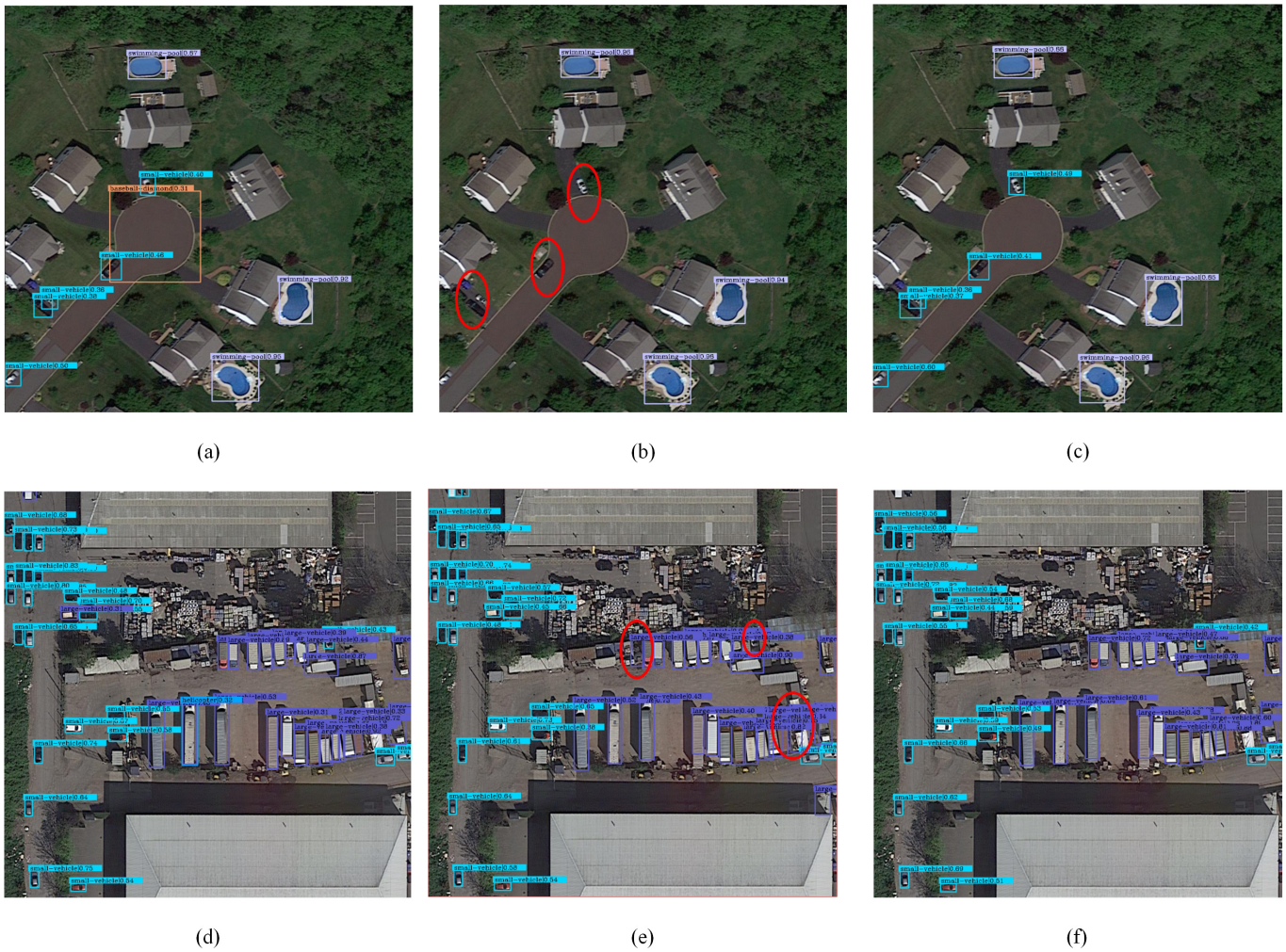


Fig. 10. Qualitative analysis of the effectiveness of the module. (a) and (d) Raw model only trained with ground-truth supervision and serves as the baseline, (b) and (e) model after normal distillation without dynamic selection, and (c) and (f) model after dynamic ISD. The red circles indicate the poor performance of the normal distillation method in cluttered and complex scenes.

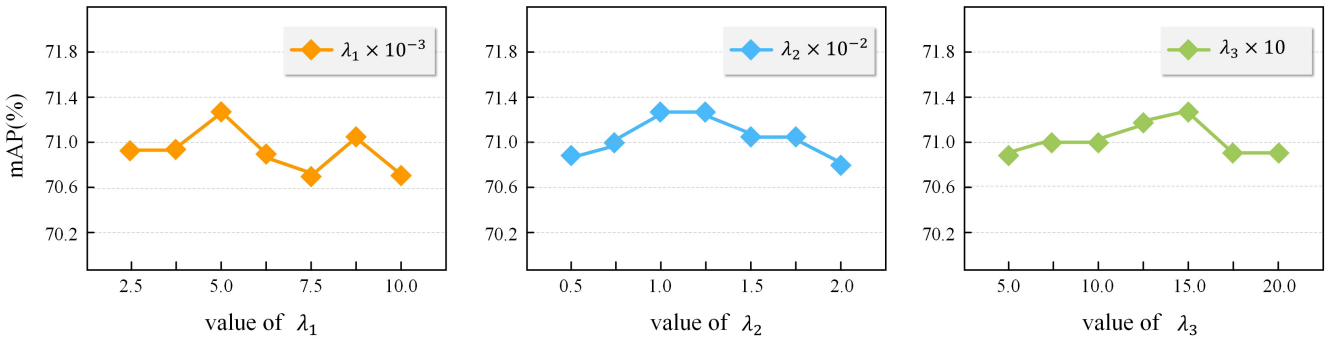


Fig. 11. Hyperparameter sensitivity study of λ_1, λ_2, and λ_3 with DKD-RetinaNet-18 on the DOTA dataset.

effects to student. Fig. 10(c) shows a network selected and distilled by dynamic modules. It achieves a good performance by taking the teacher's essence and discarding the teacher's dross.

In view of the hyperparameter threshold factor φ, it is discussed in Table V. The value of φ is directly proportional to the degree of seriousness against the teacher. We find that the influence of φ on detection performance is relatively robust. Regardless of its value, the final result is better than undifferentiated imitation.

c) Effect of training-status-aware loss-based shortest edge: After applying the loss function based on the shortest edge, as shown in Table I, the performance of the student model has been further improved. In order to improve the attention to the hard regression samples, we designed a weight w_i, which increases monotonously with the location

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TABLE VIII
DISTILL THE DIFFERENT MODELS WITH THE PROPOSED DKD METHOD ON THE NWPU VHR-10 DATASET

Base model	method	Plane	Ship	ST	BD	TC	BC	GTF	Harbor	Bridge	Vehicle	mAP	FPS
ATSS [6]	tea	99.9	92.5	92.7	96.7	97.1	92.1	99.6	91.8	73.7	98.3	93.4	18.4
	stu	98.9	87	80.9	94.4	88	75.1	98	82.3	68.3	89.2	86.2	34.9
	DKD_stu	99.8	88.1	84.5	96.6	94.8	82.7	99.6	93.6	79.8	96.7	91.6(**+5.4**)	34.3
Faster R-CNN [3]	tea	99.1	90.8	90.6	95.2	90.8	90.6	96.9	87.8	72.5	94.1	90.8	18.6
	stu	99.9	91.4	75.9	96.6	79.1	75.2	98.1	82.8	57.2	91.2	84.7	34.6
	DKD_stu	99.9	88	92.7	96.3	78.3	84.6	98.5	90.8	77.4	90.6	89.7(**+5.0**)	33.9
FCOS [5]	tea	99.8	86	82.5	96.4	92.4	96.9	98.3	83.3	87.3	93.7	91.7	18.2
	stu	99.8	87	69.5	96	89.8	72.5	98.9	89.5	72.3	87.6	86.3	36.7
	DKD_stu	98.4	88.3	77	96.7	87.5	74.4	97.7	95.6	92.1	94	90.2(**+3.9**)	34.6
RetinaNet [4]	tea	97.8	86.6	81.4	96.8	94.6	92.5	97.3	91.7	73.3	95.6	90.8	18.8
	stu	98.9	89.4	85.9	96.1	78.1	73.5	99.2	88	59.8	88.7	85.7	36.4
	DKD_stu	97.5	84.4	81.5	95.8	91.3	89	97.6	89.1	76.1	96.6	89.9(**+4.2**)	35.9

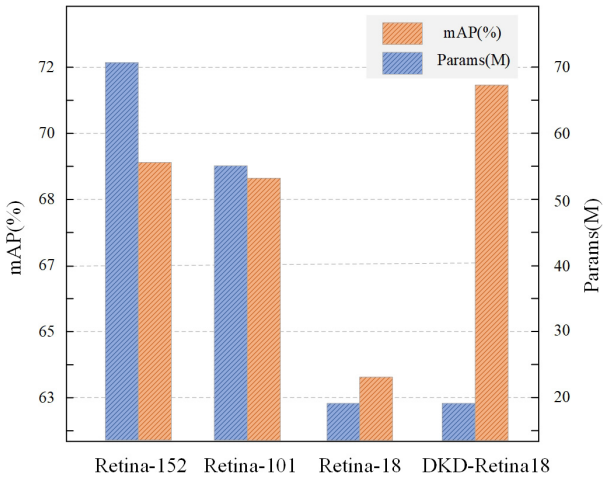


Fig. 12. Comparison the efficiency between the RetinaNet series networks with different backbones and RetinaNet-18 after distillation on the DOTA dataset.

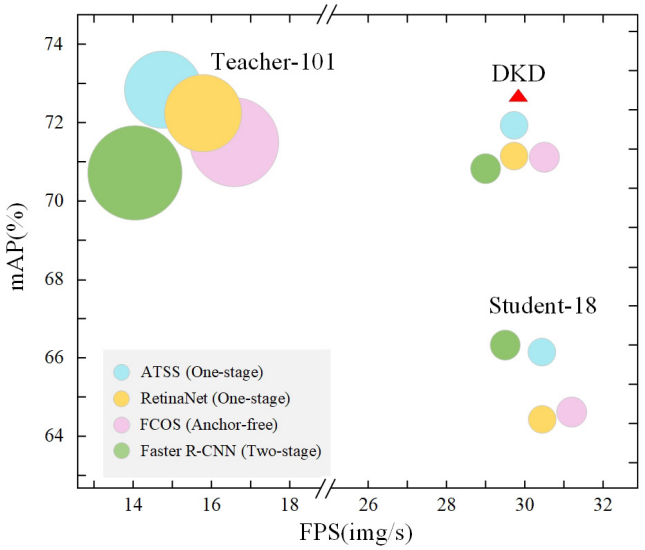


Fig. 13. Distillation effect of our DKD on typical detectors on the DOTA dataset.

difficulty. Table VI lists the argument of different weight generation methods. On the DOTA dataset, experiments show that the performance of exponential increasing function is better than that of linear increasing function.

d) Combination: The combined distillation of all the above modules has achieved the best effect, which is 1.07 higher than the best single module. In the beginning, raw student is fast and simple. However, compared with the teacher, there is a huge nine-point gap in accuracy. Our distillation framework DKD achieves a significant boost: after DKD, student gets almost the same accuracy as teacher, while the efficiency still keeps the original low cost. Furthermore, Fig. 11 shows the robustness of each hyperparameter.

3) Analysis the Benefits of KD: Figs. 14 and 15 show the qualitative results on the gain from our DKD framework on the DOTA and NWPU VHR-10 datasets. Given the same images, the top row is the RetinaNet-18's output without distillation, and the bottom is the output of RetinaNet-18 after DKD.

More Reliable Classification: As shown in Fig. 14(c), the raw student confused a large number of ST as RAs. However, the distilled student correctly classified all the objects in the image.

Less Error for Small and Large Aspect Ratio Objects: As shown in Fig. 14(d) and (f), the raw student predicted a rather inaccurate location of the bridge, harbor, and ship. While the distilled student learned better location information from the teacher. This improvement comes from the addition of the training-status-aware loss based on shortest edge.

More Accuracy for Cluttered Arranged Objects in Complex Scene: As shown in Fig. 14(f) and (g), the raw student missed the dense ships and repeatedly predicted the large vehicle. However, the student after DKD successfully solved this problem. The phenomenon shows the two functions of the DKD framework: 1) DKD makes student to have better processing capabilities for objects that appear in complex and cluttered scenes and it avoids repeated detection and 2) DKD can also improve the discriminative ability of the network. It gives higher confidence to the predicted objects so that it can be retained after nms.

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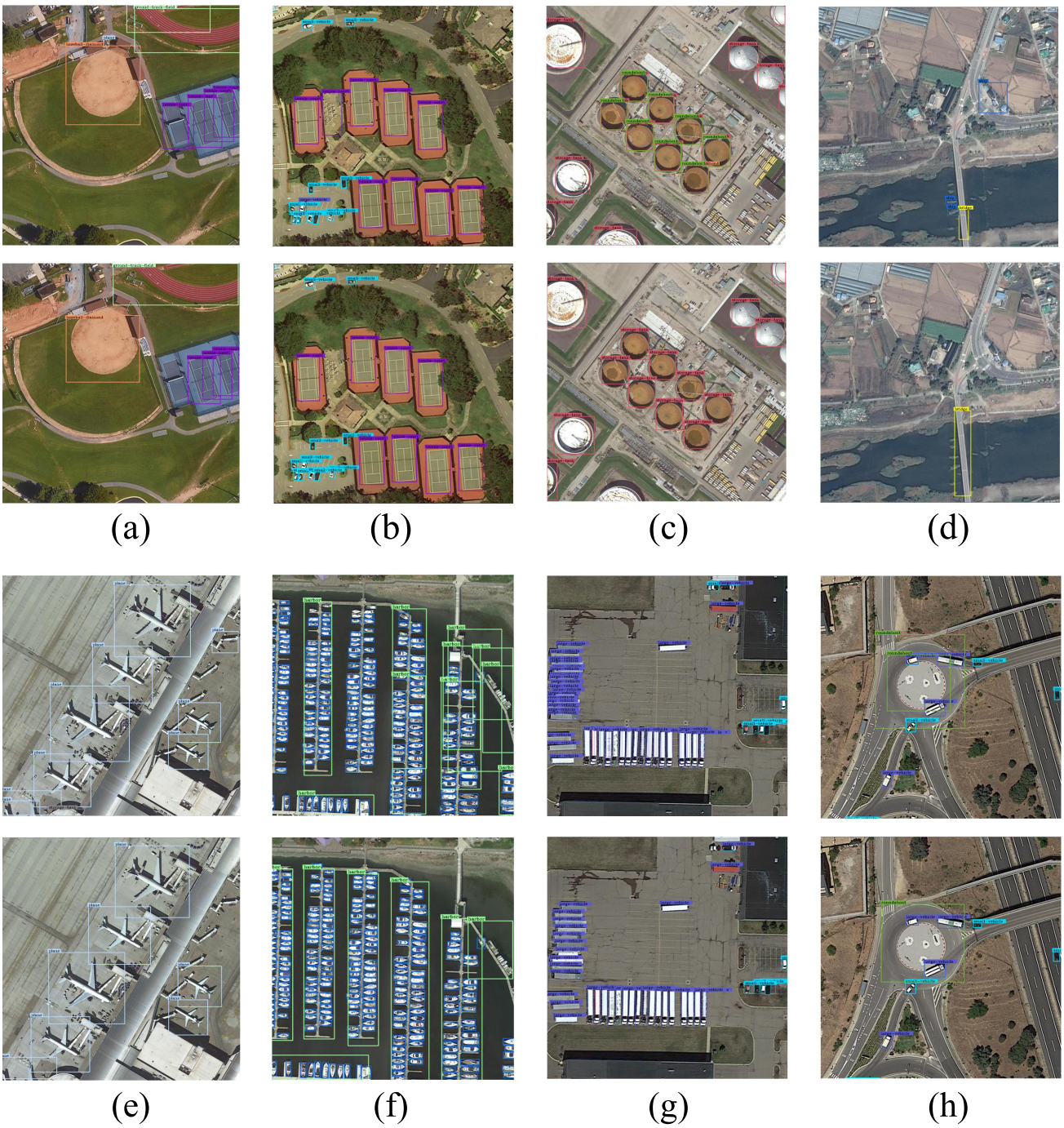


Fig. 14. Qualitative analysis of the gain in distillation learning on the DOTA dataset. Only the bounding box with confidence greater than 0.3 will be visualized. The two rows are a group. (Top) Raw student as baseline. (Bottom) Student who has passed DKD framework. (a) Baseball-diamond and ground-track field. (b) TC. (c) ST. (d) Bridge. (e) Plan. (f) Harbor and ship. (g) Large vehicle and small vehicle. (h) RA.

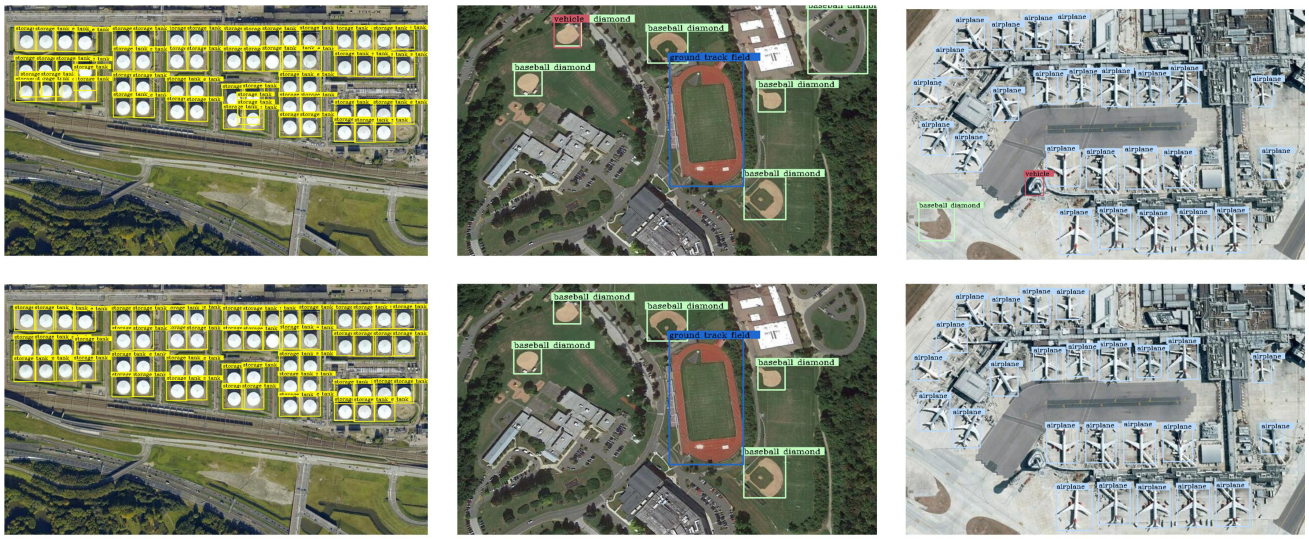
4) Efficiency Analysis: For the refinement of detectors using KD, the improvement of speed and efficiency is due to a lighter backbone network. Because the additional distillation module only participates in the training process, DKD is cost-free in the inference stage. As shown in Fig. 13, DKD improves the accuracy of the small model after distillation and keeps the original low consumption. Table VII and Fig. 12 compare the efficiency of the RetinaNet series networks with different backbone and RetinaNet-18 after DKD. We report the inference times in chips with a size of 1024.

Table VII can be seen that compared with RetinaNet-152, using the network after DKD distillation, the inference speed is increased by nearly three times, the parameter amount is reduced by nearly four times, and FLOPs is reduced by more than two times. In addition, the distilled RetinaNet-18 achieved an accuracy exceeding RetinaNet-152.

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TABLE IX
DISTILL THE DIFFERENT MODELS WITH THE PROPOSED DKD METHOD ON THE DOTA DATASET

Base model	method	Plane	BD	Bridge	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	Harbor	SP	HC	mAP	FPS
ATSS [6]	tea	88.7	78.85	53.85	61.34	80.34	68.9	83.12	90.73	80.82	84.17	51.25	65.51	75.57	79.88	53.53	73.1	14
	stu	86.86	65.18	45.23	56.18	79.2	64.33	78.27	90.7	70.41	79.34	38.53	56.66	69.34	71.35	41.88	66.23	29
	DKD_stu	87.55	76.72	49.62	63.81	80.66	67.61	78.96	90.65	77.64	82.96	48.81	65.33	73.77	77.49	51.48	71.53(**+5.3**)	29
Faster R-CNN [3]	tea	89.19	81.39	52.06	54.08	78.98	65.99	78.44	90.71	76.73	84.78	48.57	63.28	74.53	80.59	48.08	71.16	14
	stu	85.77	67.38	40.39	52.26	76.44	63.1	77.34	90.68	66.72	78.93	39.29	54.26	64.82	69.87	38.64	64.39	28
	DKD_stu	88.18	77.35	48.31	57.88	78.97	67.49	78.31	90.76	76.74	81.79	47.32	66.57	72.52	79.04	44.92	70.41(**+6.02**)	28
FCOS [5]	tea	87.48	81.17	55.46	61.21	77.22	66.57	78.7	90.15	78.64	82.06	49.91	61.67	75.26	70.82	54.16	71.36	15
	stu	86.27	71.52	46.93	57.8	72.62	65.04	77.76	90.67	69.89	75.98	45.28	65.39	69.85	66.53	44.94	66.49	29
	DKD_stu	87.54	77.74	51.58	58.13	77.64	66.55	78.17	90.79	77.85	81.71	46.81	62.56	72.46	70.98	50.14	70.04(**+3.55**)	29
RetinaNet [4]	tea	88.91	83.10	51.57	62.59	72.48	72.57	78.32	90.39	80.91	79.78	53.75	67.77	74.44	71.43	58.37	72.42	15
	stu	86.64	71.61	39.19	51.27	71.64	64.52	77.02	89.66	66.61	71.74	40.55	55.38	63.76	69.02	36.80	63.69	28
	DKD_stu	89.48	81.48	46.38	60.52	76.25	64.18	78.36	90.79	78.60	78.31	53.12	65.06	73.05	74.11	59.98	71.31(**+7.62**)	28



(a) (b) (c)

Fig. 15. Qualitative analysis of the gain in distillation learning on the NWPU VHR-10 dataset. (Top) Raw student as a baseline. (Bottom) Student who has passed the DKD framework. (a)–(c) Three groups of visualizations before and after DKD framework.

5) Versatility of Our Method: We prove the versatility of DKD from two perspectives. First, DKD was used to distill typical single-stage (Retina/ATSS), two-stage (Faster R-CNN), and even anchor-free (FCOS) detectors on the DOTA and NWPU VHR-10 datasets. Second, DKD was used in different types of backbone networks, including ResNet and HRNet.

For distilling different kinds of typical detectors, the distillation result is shown in Tables VIII and IX. With regard to the two-stage network, because the RPN output of teachers and students has different regions of interest, we add a region suggestion sharing mechanism to align the prediction instances of the teacher and student detection heads. For the anchor-free network, we did not add the training-status-aware loss based on the shortest edge in the distillation process. The experiment of Tables VIII and IX uses RetinaNet-101 as the teacher and RetinaNet-18 as the student. Combined with Figs. 14 and 15, it can be seen that DKD has achieved excellent performance for different types of detectors.

TABLE X
DISTILL THE TYPICAL DETECTOR RETINANET BASED ON HRNET WITH THE PROPOSED DKD ON THE DOTA DATASET

Method	mAP
Teacher-hrnet-w40	73.44
Student- hrnet-w18	67.38
+ DKD	71.69 (+4.31)

For distilling the detector based on different backbones, the distillation result is shown in Table X. In the experiment, RetinaNet is still used as a detector and evaluated on the DOTA dataset. However, the backbone of the teacher network is replaced by HRNetv2p-w40 [43], and the student network is replaced by HRNetv2p-w18. The former is marked as

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TABLE XI
COMPARISON WITH SOTA DISTILLATION METHODS FOR DETECTORS ON THE DOTA DATASET

Model	Plane	BD	Bridge	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	Harbor	SP	HC	mAP
teacher	88.91	83.10	51.57	62.59	72.48	72.57	78.32	90.39	80.91	79.78	53.75	67.77	74.44	71.43	58.37	72.42
student	86.64	71.61	39.19	51.27	71.64	64.52	77.02	89.66	66.61	71.74	40.55	55.38	63.76	69.02	36.80	63.69
+Wang [26]	88.43	75.97	44.49	56.28	72.91	63.55	74.90	90.81	76.74	72.14	48.40	61.64	70.03	69.04	45.53	67.39
+Sun [27]	89.12	76.50	**46.52**	59.95	73.87	**66.22**	**78.53**	90.76	78.59	75.67	43.67	**66.65**	71.20	72.05	45.29	68.97
+ours	**89.48**	**81.48**	46.38	**60.52**	**76.25**	64.18	78.36	**90.79**	**78.60**	**78.31**	**53.12**	65.06	**73.05**	**74.11**	**59.98**	**71.31**

TABLE XII
COMPARISON BETWEEN THE DISTILLED DETECTORS AND ADVANCED DETECTORS ON THE DOTA DATASET

Method	backbone	MS	Plane	BD	Bridge	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	Harbor	SP	HC	mAP	FPS
R-FCN [51]	ResNet101		81.01	58.96	31.64	58.97	49.77	45.04	49.29	68.99	52.07	67.42	41.83	51.44	45.15	53.3	33.89	52.58	9
Def FR-H [52]	ResNet101	✓	86.53	77.54	42.7	64.43	67.6	63.64	77.86	90.33	77.82	75.36	52.12	56.79	68.92	62.04	54.92	67.91	4
FMSSD [8]	VGG16		89.11	81.51	48.22	67.94	69.23	73.56	76.87	90.71	82.67	73.33	52.65	67.52	72.37	80.57	60.15	72.43	16
ICN [53]	ResNet101	✓	90.00	77.70	53.40	73.30	73.50	65.00	78.20	90.80	79.10	84.80	57.20	62.10	73.50	70.20	58.10	72.50	——
IoU-Adaptive [54]	ResNet101		88.62	80.22	53.18	66.97	76.3	72.59	84.07	90.66	80.95	76.24	57.12	66.65	84.08	66.36	56.85	72.7	1
FADet [55]	ResNet101	✓	90.15	78.6	51.92	75.23	73.6	71.27	81.41	90.85	83.94	84.77	58.91	65.65	76.92	79.36	68.17	75.38	——
Mask OBB [56]	ResNeXt-101	✓	89.69	87.07	58.51	72.04	78.21	71.47	85.2	89.55	84.71	86.76	54.38	70.21	78.98	77.46	70.4	76.98	——
DKD-SRAFNet [7]	ResNet18		86.67	74.65	55.10	59.31	77.28	66.18	78.67	89.53	77.69	83.62	46.15	61.75	75.76	70.41	54.20	70.47	21
DKD-CGNet [57]	ResNet18		87.93	85.14	56.27	72.54	72.01	80.94	88.36	91.25	84.32	82.52	62.11	69.19	78.25	75.31	67.21	76.89	25

"Teacher-hrnet-w40," and the latter is marked as "Student-hrnet-w18." As shown in Table X, the teacher network and the student network bring 73.44% mAP and 67.38% mAP accuracies, respectively, with a gap of 6.06% mAP. Our distillation framework DKD achieves a significant boost: after DKD, student network gains 4.31% mAP of improvement. Overall, DKD is very robust and universal.

6) Comparisons With Other Methods: In this section, we compare the proposed DKD with the state-of-the-art methods from the perspectives of distillation and detection on the DOTA dataset.

Table XI is based on RetinaNet as the basic network. RetinaNet with the backbone of ResNet-101 is used as the teacher network and RetinaNet with the ResNet-18 is used as the student network. The experiments are carried out on the DOTA dataset. Compared with existing distillation frameworks, DKD enables typical detection networks to achieve very superior performance in the field of remote sensing under the premise of the same time cost.

Table XII shows the comparison between the distilled model and advanced detectors on the DOTA dataset. "MS" means using multiscale training and testing which can further improve the performance. In fact, DKD as a distillation framework is used to lighten the existing detectors. However, in order to further prove the effectiveness and feasibility of our framework in remote sensing field, we list two networks after distillation in this section. We compare their results with advanced detectors specifically designed for remote sensing.

V. CONCLUSION

In this article, we proposed a general and effective KD framework called DKD to lightweight the state-of-the-art detector. First, our framework leverages the dynamic GD module to capture spatial unique importance for multiscale features imitation. GD automatically finds valuable regions from the foreground and background to distillation, avoiding ignoring the potential geographical spatial relationship. Second, we proposed a dynamic ISD module to give the student network the ability of self-judgment through the magnitude of detection loss. It can discriminately imitate masses of instances generated by teachers in complex and cluttered scenes. Third, toward more accurate handling of hard samples in regression, a training-status-aware loss based on the shortest edge is tailored to guide student mine knowledge about objects with large aspect ratio or small size. Finally, extensive experiments are conducted to show the effectiveness and universality of DKD framework. The detection results on the DOTA and NWPU VHR-10 datasets illustrate that our method is suitable for typical single-stage, two-stage, and even anchor-free detectors and show state-of the-art performance.

In the future, we will be full of expectations for the following three aspects of exploration. First, we hope to get rid of the distillation paradigm of the freezing teacher and obtain a soft teacher who can grow in training [22], [23]. Second, the multiscale knowledge of paired teacher–student needs to be further utilized. The cross-layer or fused multiscale features that the student learn from the teacher may bring new growth points to distillation [24], [25]. Third, building a universal distillation framework for remote sensing images with oriented bounding boxes [41], [42], [50] is also a very challenging and meaningful work. In general, we will work to boost the distilled detector's performance and further improve the efficiency of the compression mode.

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